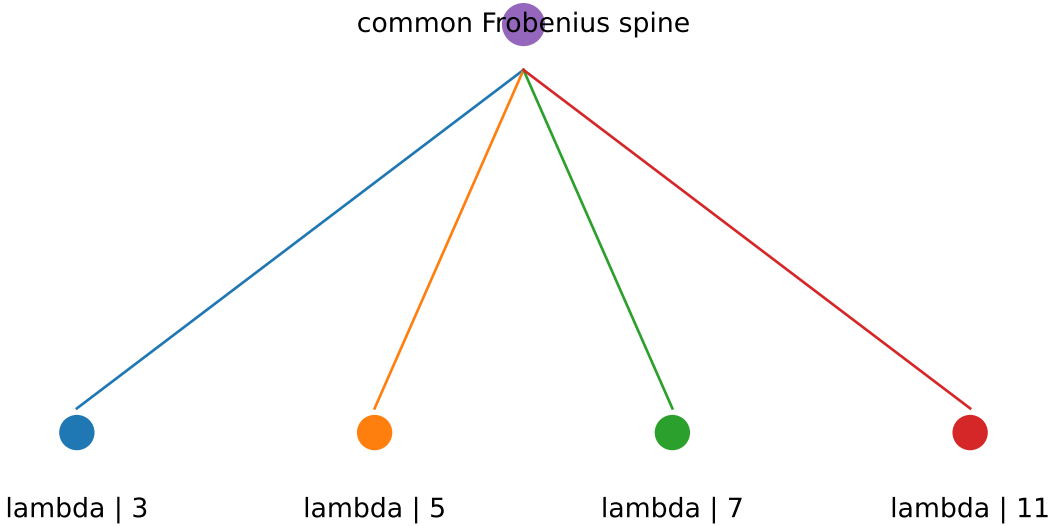Local coefficient-prime members glued through one algebraic spine